

A meta-analysis of correction effects in science-relevant misinformation

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Scientifically relevant misinformation, defined as false claims concerning a scientific measurement procedure or scientific evidence, regardless of the author's intent, is illustrated by the fiction that the coronavirus disease 2019 vaccine contained microchips to track citizens. Updating science-relevant misinformation after a correction can be challenging, and little is known about what theoretical factors can influence the correction. Here this meta-analysis examined 245 effect sizes (that is, k , obtained from 75 reports; $N = 53,320$), which showed that attempts to debunk science-relevant misinformation were, on average, not successful ($d = 0.11$, $P = 0.142$, 95% confidence interval -0.04 to 0.26). However, corrections were more successful when the initial science-relevant belief concerned negative topics and domains other than health. Corrections fared better when recipients were likely familiar with both sides of the issue ahead of the study and when the issue was not politically polarized.

Unfounded misinformation about, for example, the false association between the coronavirus disease 2019 (COVID-19) pandemic and the rollout of 5G cellular tower networks^{1,2} requires correction because it misleads citizens and can undermine their wellbeing^{3–6}. Therefore, it is important to understand the efficacy of corrections of science-relevant misinformation using different methods, including research synthesis. In this Article, we define misinformation as ‘false information’ (see also refs. 7–10), including ‘information considered incorrect based on the best available evidence from relevant experts at the time’¹¹, ‘fabricated information that mimics news media content in form’¹², ‘fictitious misinformation concerning a scientific measurement procedure or scientific evidence for example’¹³, and scientific findings that have been proven false^{14,15}. Specifically, we examined science-relevant misinformation, defined as false claims concerning a scientific measurement procedure or scientific evidence, regardless of the author's intent, as opposed to, for example, political misinformation. We considered research on media campaigns to correct vaccine misconceptions^{16,17}, correction of false research reports about the impact of personality traits on the performance of firefighters¹⁸, and correction of fictitious fake news coverage about scientific issues^{16,19}. All in all, we analysed science-relevant misinformation, which we define as false claims attributed to scientific methods or scientists in areas such as social science, climate change or health. For example, the claim that the COVID-19 vaccine decreases

fertility is pseudo-medical and thus comprises a science-relevant claim. This type of misinformation excludes non-scientific information about the same topics, such as a false claim that a politician made a particular statement about the COVID-19 vaccine or that another politician has refused to vaccinate.

Our meta-analysis was driven by theoretical explanations that suggest moderators of the impact of corrections as well as the initial misinformation. Specifically, we investigated how the nature of the misinformation, the correction and the recipient affect the correction. These factors, which are shown in Fig. 1, involve the valence of misinformation (that is, negative versus neutral) and the use of detailed (versus succinct) corrections. We also considered the attitudinal congeniality of the corrections (that is, congenial versus mixed/uncongenial) and issue polarization (that is, polarizing versus not polarizing) among the recipients.

The innovation of our meta-analysis was to synthesize the impact of science-relevant misinformation and its correction. Specifically, we are interested in two research questions. First, to what degree can the public update science-relevant misinformation after a correction? Second, what theoretical factors (that is, negative misinformation, detailed correction, attitudinal congeniality of the correction, and issue polarization) influence the impact of corrections? To address these questions, we synthesized reports of experiments studying the correction of science-relevant misinformation. We included corrections of

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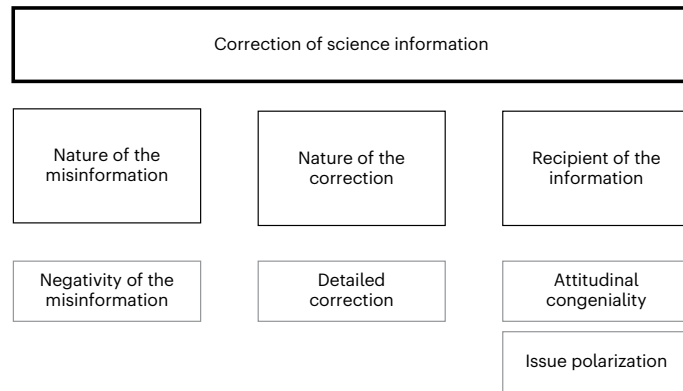


Fig. 1 | Theoretical factors related to the correction of science-relevant misinformation.

misconceptions that circulate in the real world such as misinformation about climate change, genetically modified organisms, COVID-19 and vaccines^{20,21}. We also included corrections of fictitious misinformation concerning a scientific measurement procedure or scientific evidence (for example, ref. 13) and corrections of scientific findings that have been proven false (for example, refs. 14,15).

Six prior meta-analyses have examined the persistence of misinformation in news and reports^{22–27}. However, none had the same goals as ours. Three of the prior meta-analyses did not concentrate on science-relevant information. Chan et al.²² meta-analysed eight reports (k of effect sizes, 52) of research that used fictitious social and political news as experimental materials. Two other meta-analyses concentrated on social and political news such as restaurant rumours and news about political events^{24,25}. The three that did consider correction of science-relevant misinformation^{23,26,27} covered either climate change or health but not science-relevant information more generally. The first assessed five effect sizes concerning corrections of vaping misinformation²³, the second synthesized 15 effect sizes reflecting corrections of climate-change or health misinformation²⁶, and the third examined 24 effect sizes about the impact of a source's reliability ratings on correction of health misinformation on social media²⁷. In sum, each of these meta-analyses considered a small number of effect sizes (that is, $k = 5–24$) without examining the moderators we examined or our general question (Fig. 1). These moderators are discussed in turn.

The misinformation

An important consideration about the impact of corrections is the valence of the misinformation, particularly whether the topic can arouse negative emotions in an audience. Given that much of the scientific information disseminated to the public is upsetting^{28–31}, we first wondered if negative science-relevant misinformation is easier or more difficult to correct than their neutral counterpart. For example, the alleged side effects of infertility or autoimmune diseases following vaccination against human papilloma virus (HPV)^{32,33} can elicit fear or sadness in an audience^{33–37}. These negative emotional implications may affect corrections of this information, although the direction of influence is debatable. On the one hand, negative information may elicit more attention and more thorough processing^{38,39}, which may, in turn, increase the persistence of negative (versus neutral) misinformation. On the other hand, people are more likely to hold beliefs that make them feel good about themselves, their future or the world more generally^{40–42}. As a result, negative (versus neutral) misinformation may be easier to correct because doing so may improve a person's mood.

The correction

Different correction factors may also affect the correction of science-relevant misinformation. According to the notion of mental

models, people's ability to discard a model built on misinformation depends on the strength of that model and the ability of the correction to promote a new model^{43,44}. Accordingly, detailed corrections and causal explanations can change prior models and make correcting misinformation (see also refs. 45,46). Yet, detailed corrections and explanations may have the ironic effect of strengthening misinformation persistence²². For example, detailed corrections and elaborate explanations of the misinformation may remind the audience of the misinformation^{47,48}. As no prior meta-analysis has assessed the influence of detailed (versus succinct) corrections of science-relevant misinformation, we attempted to fill this void by analysing this moderator.

The recipient

Whether a correction is congenial to the recipient's attitudes or beliefs may also affect the success of corrections of science-relevant information⁴⁹. A 2020 meta-analytic review of political misinformation²⁵ found that corrections were more efficacious when they were congruent with recipients' attitudes than when they were not^{50–52}. However, the evidence about the impact of congeniality is not monolithic. For example, receiving misleading headlines congenial to recipients' political ideology does not impair recipients' ability to distinguish true from false information^{7,53} or their motivation to share only accurate information with their social networks¹⁰. Furthermore, we considered if the topic is politically polarized. If people engage in motivated reasoning to protect their political identity^{40,54–56}, for example, corrections should have lesser effect when the issues are politically polarized. However, corrections have been shown to work for polarized misinformation as well⁵⁷. Our meta-analysis thus estimated the impact of both congeniality and political polarization on corrections of science-relevant information.

The present study

We synthesized a large body of experimental evidence comprising 61 published experiments, 5 working papers, 2 theses and 7 unpublished datasets (total number of reports, 75; k of effect sizes, 245). We examined the moderating effects of factors related to the misinformation (that is, negativity), the correction (that is, detailed correction), the recipient (that is, attitudinal congeniality and issue polarization), control factors (that is, domain, fictitious issue, likely familiarity and in-person correction) and report and methodological characteristics (that is, study sample, lab context and method of effect-size calculation). We also assessed the misinformation effect to ensure that our moderators reflected the impact of the correction rather than the impact of the misinformation. All procedures are detailed in Methods.

In addition to the theoretical moderators included in Fig. 1, we controlled for factors that could vary across experiments. We controlled for the domain of the misinformation (that is, political, health, environment and others), whether the misinformation was fictitious⁵⁸, whether the audience was likely familiar with the topic and whether the correction was delivered in person^{59,60}. We also considered two report characteristics and one methodological characteristic that might affect corrections, including study sample (that is, the United States versus other countries), lab context (that is, laboratory versus online) and method of effect-size calculation (that is, between subjects versus within subjects). We chose these control characteristics on the basis of a review of prior meta-analyses^{22,26} and the availability of these details in our sample of reports.

Analytic procedures

We estimated Hedges' d for the effect of the correction with adjustments to minimize small sample bias and on the basis of either between-subjects or within-subjects variances depending on the procedures of the included studies. We subtracted the mean belief or attitude rating after a correction was introduced from the mean rating before a correction or in a control condition. We used four add-on packages for the statistical software R version 4.0.5: robumeta

Table 1 | Descriptive statistics of the included reports

	Mean (s.d.)	k
Sample size	217.63 (352.18)	245
Percentages of females	58.53 (13.99)	154
Percentages of males	40.20 (14.20)	139
Age	36.70 (10.49)	107
Country		
United States (%)	67%	163
Other countries (%)	33%	80

version 2.0 (refs. 61,62), metafor version 3.9.9 (ref. 63), puniform version 0.2.5 (ref. 64) and weightr version 2.0.2 (ref. 65) to assess publication/inclusion bias and analyse the mean effect sizes using robust variance estimation (RVE) methods^{66–68}. We also used JASP version 0.16.3 (ref. 69), an open-source statistics program, to conduct bias analyses with Bayesian methods. We calculated the I^2 statistic (that is, the percentage of total variation across experimental conditions due to random heterogeneity), which controls for k and indicates the percentage of total variation across experimental conditions that is due to true heterogeneity rather than sampling error^{70,71}. Furthermore, we performed meta-regression analyses of our debunking effect sizes with the moderators in Fig. 1 introduced as predictors and repeated this analysis with the misinformation effect as an outcome and later as a covariate. Because debunking effects involved both correction and the reverse of misinformation persistence, these analyses utilized RVE to account for the statistical dependence between the two, which was estimated at 0.53 (that is, ρ). A separate sensitivity analysis, as recommended by Hedges et al.⁶⁸, was performed to confirm that the selected ρ estimate was appropriate ($\tau^2 = 0.40$ for ρ ranging from 0 to 1). The pre-registration materials and data and code repositories are available at <https://osf.io/vkygw/> (the pre-registration was done before the last round of database update in mid-2022).

Results

We included 75 research reports and 245 independent effect sizes (for the included data, see Supplementary Table 1). According to a review by two authors (Methods), these conditions met inclusion criteria in that they (1) provided a false claim concerning a scientific measurement procedure or scientific evidence, (2) had measures of participants' beliefs or attitudes consistent with the misinformation addressed by the correction and (3) had a baseline or control group. Also, studies were eligible when (4) the misinformation was initially asserted to be true or was known to participants before the study and was later corrected (Methods). Table 1 reports descriptive statistics of the included reports. The number of participants in the synthesis ranged from 7 to 3,142, their mean age was 37 years (standard deviation (s.d.) 10.49) and about 60% of them were female. Participants included university students, graduate students and adults from the community recruited via online survey platforms, such as Amazon Mechanical Turk and Prolific.

Overall effects, heterogeneity and bias

Although the misinformation effect was large ($d = 0.58$, $P < 0.001$, 95% confidence interval (CI) 0.36 to 0.80), the debunking effect was not significant ($d = 0.11$, $P = 0.142$, 95% CI -0.04 to 0.26). The heterogeneity analyses showed high I^2 statistics (misinformation 99.03% and debunking 97.19%), suggesting systematic variability across conditions in addition to sampling error or multiple populations of effects⁷¹. We also conducted extensive analyses of inclusion bias, which appear in Table 2 (see details of the bias analyses in Methods). These analyses revealed no consistent evidence of bias in the dataset.

Table 2 | Summary of bias analyses

Bias analysis	Result with the outliers	Result without the outliers	Indication of bias
Contoured-enhanced funnel plot with the trim-and-fill method ^{118–120}	LO: 0 estimated records filled on the right RO: 0 estimated record filled on the right	LO: 0 estimated records filled on the right RO: 5 estimated records filled on the right	LO: No RO: Yes, for the results with the outliers
Rank correlation test	Kendall's tau = -0.09 , $P = 0.023$	Kendall's tau = -0.09 , $P = 0.049$	Yes
Precision-effect test—precision-effect estimate with standard error ¹²¹	Small difference between the PEESE and the RVE estimate, $d_{\text{diff}} = 0.17$	Small difference between the PEESE and the RVE estimate, $d_{\text{diff}} = 0.16$	No
p -uniform test with the default “P” method set ¹²²	L.pb = -6.49 , $P = 1.000$	L.pb = -6.21 , $P = 1.000$	No
Three-parameter selection model ¹²³	$b = -0.07$, SE = 0.07, $P = 0.249$	$b = -0.02$, SE = 0.06, $P = 0.795$	No
Meta-regression test of publication type	Working paper: $b = 0.14$, SE = 0.12, $P = 0.231$ Dissertation/Thesis: $b = 0.09$, SE = 0.13, $P = 0.497$ Unpublished data: $b = -0.04$, SE = 0.10, $P = 0.677$	Working paper: $b = 0.11$, SE = 0.10, $P = 0.304$ Dissertation/Thesis: $b = 0.06$, SE = 0.12, $P = 0.624$ Unpublished data: $b = -0.07$, SE = 0.08, $P = 0.368$	No
Weight-function models ⁶⁵	Log-likelihood ratio was significant χ^2 (df = 3) = 66.61, $P < 0.001$	Log-likelihood ratio was significant χ^2 (df = 3) = 52.42, $P < 0.001$	Yes
Robust Bayesian meta-analysis ¹²⁴	BF ₁₀ = 0.043	BF ₁₀ = 0.073	No

b indicates unstandardized coefficients, SE indicates standard error, BF₁₀ indicates Bayes factor giving the evidence for H1 over H0, and P values ≥ 0.05 were concluded as no indication of the presence of bias.

Moderator analyses

Table 3 presents the results from meta-regressions, and Table 4 reports the predicted estimated mean effect sizes for each level of the categorical moderators using the identified meta-regression model. First, corrections were more effective for negative (versus neutral) misinformation ($b(60) = 0.27$, $P = .017$, 95% CI 0.05 to 0.50). Next, corrections were more successful concerned a non-polarizing (versus polarizing) issue ($b(60) = 0.54$, $P = .014$, 95% CI 0.11 to 0.96). Of note, corrections were successful irrespective of whether they were detailed (versus succinct) ($b(60) = 0.14$, $P = 0.316$, 95% CI -0.14 to 0.42) and whether they were congenial (versus mixed/uncongenial) ($b(60) = 0.30$, $P = 0.150$, 95% CI -0.11 to 0.71). As for control factors, corrections were more efficacious when the misinformation concerned other (versus health) topics ($bs(60) = 0.47 - 0.70$, $Ps = 0.007 - 0.044$, 95% CIs 0.01 to 1.21) and when recipients were likely familiar (versus unfamiliar) with the topic ($b(60) = 1.04$, $P = 0.001$, 95% CI 0.47 to 1.61). No statistical significant effect on correction was found when the issues were fictitious (versus real) ($b(60) = 0.04$, $P = 0.827$, 95% CI -0.32 to 0.40). There was no statistically significant difference in in-person (versus through media) corrections ($b(60) = 0.04$, $P = 0.845$, 95% CI -0.37 to 0.44). As shown in Table 3, these meta-regressions also controlled for whether the samples were from the United States (versus other countries), whether the studies were conducted in the lab (versus online) and whether the effect sizes were computed using the between- or within-subjects methods. Of these variables, none were statistically significant.

Table 3 | Meta-regression results

Variable	Debunking effect			Misinformation effect			Debunking effect controlling for the misinformation effect		
	<i>k</i> = 243			<i>k</i> = 42			<i>k</i> = 76 ^a		
	<i>b</i> (SE)	<i>P</i>	95% CI	<i>b</i> (SE)	<i>P</i>	95% CI	<i>b</i> (SE)	<i>P</i>	95% CI
Intercept	−0.61 (0.22)	0.007	−1.04 to −0.17	0.75 (0.29)	0.010	0.18 to 1.32	−0.43 (0.21)	0.071	−0.91 to 0.05
Nature of the misinformation									
Negative misinformation [^]	0.27 (0.11)	0.017	0.05 to 0.50	0.84 (0.38)	0.025	0.11 to 1.58	0.01 (0.21)	0.975	−0.47 to 0.48
Nature of the correction									
Detailed correction [^]	0.14 (0.14)	0.316	−0.14 to 0.42	−0.32 (0.18)	0.077	−0.68 to 0.03	0.06 (0.14)	0.670	−0.27 to 0.39
Recipients of the misinformation									
Attitudinal congeniality of the correction [^]	0.30 (0.21)	0.150	−0.11 to 0.71	-	-	-	-	-	-
Issue polarization [^]	−0.54 (0.21)	0.014	−0.96 to −0.11	−2.05 (0.38)	0.000	−2.79 to −1.31	−0.07 (0.21)	0.753	−0.56 to 0.42
Control factor:									
Domain of misinformation: political [^]	0.91 (0.23)	0.000	0.44 to 1.37	-	-	-	-	-	-
Domain of misinformation: health [^]	−0.7 (0.25)	0.007	−1.21 to −0.2	-	-	-	-	-	-
Domain of misinformation: environment [^]	−0.47 (0.23)	0.044	−0.93 to −0.01	-	-	-	-	-	-
Fictitious issue [^]	0.04 (0.18)	0.827	−0.32 to 0.40	−0.02 (0.27)	0.939	−0.54 to 0.5	−0.29 (0.16)	0.109	−0.65 to 0.08
Likely familiarity with the topic [^]	1.04 (0.28)	0.001	0.47 to 1.61	0.62 (0.28)	0.025	0.08 to 1.17	0.36 (0.24)	0.161	−0.18 to 0.91
In-person correction [^]	0.04 (0.2)	0.845	−0.37 to 0.44	−0.26 (0.24)	0.287	−0.74 to 0.22	0.86 (0.15)	0.000	0.52 to 1.19
Report and methodological characteristics									
Study sample [^]	−0.25 (0.15)	0.105	−0.55 to 0.05	−0.08 (0.48)	0.864	−1.02 to 0.85	1.01 (0.21)	0.001	0.53 to 1.49
Lab context [^]	0.38 (0.31)	0.222	−0.23 to 0.99	-	-	-	0.55 (0.32)	0.123	−0.18 to 1.28
Method of the effect-size calculation [^]	0.14 (0.13)	0.277	−0.12 to 0.41	-	-	-	−0.18 (0.12)	0.189	−0.46 to 0.11
Misinformation effects	-	-	-	-	-	-	0.37 (0.14)	0.025	0.06 to 0.69

A caret indicates a categorical variable. Unstandardized coefficients and standard errors in parentheses. RVE was used for the debunking effects, and mixed-effect estimation was used for the misinformation effects. ^a*k* = 76 because the same misinformation effect was assigned to the correction effect and the misinformation-persistence effect in the RVE estimation.

An important consideration, however, is whether the effects of negative misinformation, issue polarization, different domains, and likely familiarity with the topic might impact the misinformation impact rather than the correction per se. Therefore, we next regressed the misinformation effect of the same moderators, an analysis that appears in the second column of Table 3. Additionally, we included the misinformation effect as a covariate and conducted a meta-regression analysis of the debunking effects with the sample available for it, which was only 76 effect sizes (see the third column of Table 3) as well as with imputation for missing misinformation effects (Supplementary Table 5).

Discussion

This meta-analysis assessed two critical questions: To what degree can the public update science-relevant misinformation after a correction? And, what theoretical factors (that is, negative misinformation, detailed correction, attitudinal congeniality of the correction and issue polarization) influence the impact of corrections? We showed that science-relevant misinformation is particularly challenging to eliminate. In fact, the correction effect we identified in this meta-analysis ($d = 0.39$, $P < .001$, 95% CI 0.28 to 0.50) is smaller than those identified in all other areas (for example, $d = 1.14$ – 1.33 , 95% CI 0.62 to 2.04 in Chan, Jones, Jamieson & Albarracín²² and $d = 0.40$ – 0.75 in Walter et al.²⁴ and Walter & Murphy^{25,26}). We also identified

conditions under which corrections are most effective, including negative misinformation and issue polarization. Both had not been examined in prior meta-analyses.

Our meta-analysis can provide insights into developing evidence-based interventions for science-relevant misinformation. Although there is a growing interest in the development of effective interventions to curb the impact of misinformation⁷², the majority of the proposed mechanisms have focused on either the impact of the misinformation or the cognitive processing of corrections. In this context, our work suggests that the correction effects are a joint function of multiple factors concerning the misinformation and the recipient of the information (Fig. 2). The findings thus provide an integrated model that can better explain the complexity of the processes at hand.

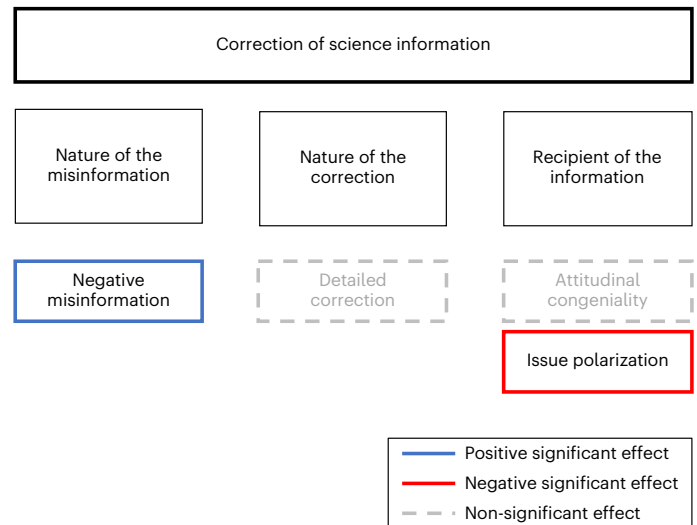
Our findings about attitudinal congeniality and issue polarization are also important in the context of discussions about different reasoning accounts of misinformation^{10,50,52,73}. The effect of attitudinal congeniality of correction seems not to be in line with recent experimental data that false headlines from media sources congenial to recipients' political ideology are perceived to be more accurate⁷⁴. However, as Traberg and van der Linden's work concerned misinformation, future should investigate if political congeniality improves the impact of corrections. Our results concerning issue polarization provide some support for the motivated reasoning^{40,56}

Table 4 | Predicted estimated mean effect sizes for levels of categorical moderators

Variable	df	<i>d</i> (SE)	<i>P</i>	95% CI
The misinformation				
Negativity of the misinformation				
Negative	17.01	0.23 (0.09)	0.016	0.05 to 0.41
Neutral	25.97	−0.04 (0.08)	0.601	−0.20 to 0.12
The correction				
Detailed correction				
Detailed	30	0.17 (0.10)	0.085	−0.02 to 0.36
Succinct	32.86	0.03 (0.08)	0.721	−0.14 to 0.20
The recipient				
Attitudinal congeniality of the correction				
Congenial	2.75	0.37 (0.21)	0.333	−0.34 to 1.08
Mixed/uncongenial	50.91	0.07 (0.06)	0.252	−0.05 to 0.19
Polarizing issue				
Yes	31.76	−0.23 (0.15)	0.140	−0.53 to 0.08
No	22.44	0.31 (0.09)	0.003	0.12 to 0.50
Control factor				
Domain of the misinformation				
Political	14.05	0.82 (0.22)	0.002	0.35 to 1.29
Health	25.02	−0.51 (0.23)	0.038	−0.03 to −0.99
Environment	19.53	−0.30 (0.19)	0.134	−0.70 to 0.10
Other	17.77	0.13 (0.12)	0.301	−0.13 to 0.38
Fictitious issue				
Fictitious	37.15	0.09 (0.07)	0.206	−0.05 to 0.23
Real	15.26	0.05 (0.16)	0.764	−0.29 to 0.38
Likely familiarity with the topic				
Yes	14.78	0.52 (0.14)	0.003	0.22 to 0.81
No	19.79	−0.52 (−0.17)	0.007	−0.16 to −0.88
In-person correction				
Yes	7.70	0.11 (0.19)	0.587	−0.34 to 0.56
No	45.76	0.07 (0.07)	0.285	−0.06 to 0.20
Report and methodological characteristics				
Study sample				
United States	41.48	0.16 (0.06)	0.008	0.04 to 0.27
Rest of the world	21.11	−0.09 (0.14)	0.513	−0.38 to 0.20
Lab context				
Lab	22.68	−0.14 (0.20)	0.49	−0.56 to 0.27
Online	20.92	0.24 (0.13)	0.086	−0.04 to 0.51
Method of effect-size calculation				
Between-subjects	43.09	0.04 (0.07)	0.61	−0.11 to 0.19
Within-subjects	23.17	0.18 (0.09)	0.062	−0.01 to 0.37

df indicates degrees of freedom with the small sample correction, *d* indicates the Cohen's *d* predicted *d*s of the meta-regression analysis, SE indicates standard errors. The predicted *d*s were estimated while keeping the covariates of negative misinformation (0.42), detailed correction (1.32), attitudinal congeniality of the correction (0.02), issue polarization (0.44), political domain of misinformation (0.18), health domain of misinformation (0.17), environmental domain of misinformation (0.20), fictitious issues (0.67), likely familiarity with the topic (0.58), in-person correction (0.11), study sample (0.33), lab context (0.57), and method of effect-size calculation (0.26) at their grand means.

and identity-based accounts^{75,76} of correcting misinformation. Corrections become less efficacious when the issue is polarizing, possibly because recipients defend themselves against identity threats and counterargue the correction. However, these accounts are difficult

**Fig. 2 | Findings of theoretical factors related to the correction of science-relevant misinformation.**

to separate from the fact that issues one agree with simply appear more valid and need to be validated with a demonstration of the impact of goals.

Our results suggest practical recommendations for undercutting the influence of science-relevant misinformation. To maximize efficacy, corrections should be accompanied with methods to reduce polarization around an issue. For example, thinking of a friend with a different political ideology can reduce affective polarization^{77,78}. Lastly, corrections are likely to be more effective when recipients are familiar with the topic. Therefore, increasing public exposure to the topics (for example, general information about a subject matter) may also maximize the impact of debunking.

Even though this meta-analysis, to our knowledge, is the most comprehensive in this area, our conclusions have limitations imposed by the existing literature. First, because little experimental work has assessed the impact of repeating the misinformation or the corrections^{79–81}, future research should address that problem. Second, no experiment measured or estimated people's understanding of the scientific process, which is one of the key factors in science communication^{82,83}. Although the level of education attained by participants can serve as a proxy for this knowledge, only one experiment reported the education level attained for each condition in their results⁸⁴. Note that a separate meta-regression ($k = 107$) with the mean education of a study as a whole included as an additional moderator revealed no statistically significant association with the debunking effect ($P = 0.995$). Future research should report educational attainment, ideally for each experimental condition, to assess whether education moderates misinformation and correction effects. Third, the I^2 statistics still showed a high proportion of random heterogeneity (that is, between-studies variability) even after controlling for our moderators. Other factors that contribute to this unexplained heterogeneity may include variability in the social environment, conditions of study administration and experimental paradigms, which may not be discernible from published results but may nonetheless affect study results. Finally, researchers should pre-register their experiments to increase the transparency of their methodologies and improve reproducibility. Direct replications using shared experimental paradigms may overcome the limitations of single experiments and control for the differences in the studies included in a meta-analysis⁸⁵. Taken together, the meta-analytic and replication efforts should provide complementary evidence about how to best protect populations from the dangers of pseudo-scientific misinformation.

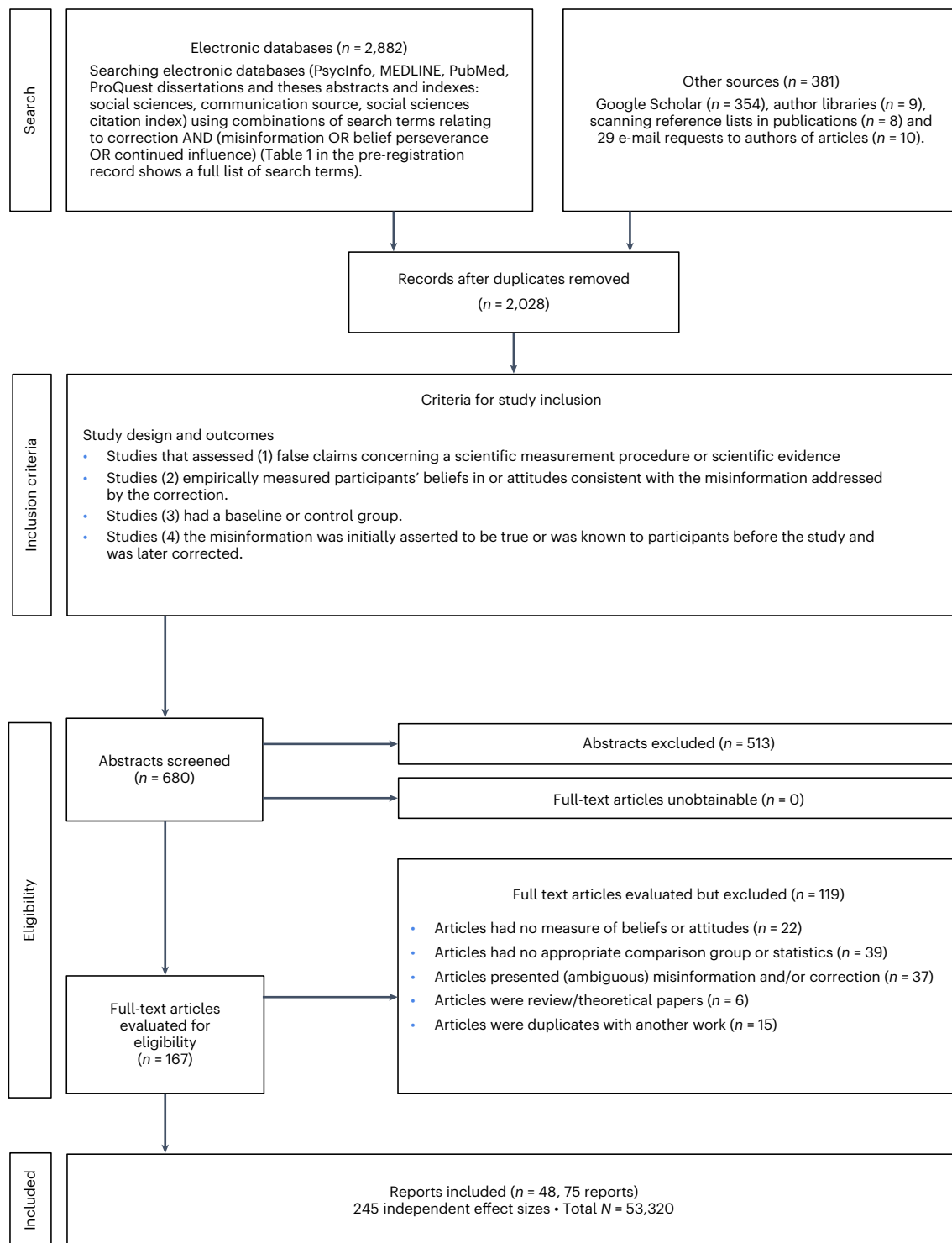


Fig. 3 | Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram.

Methods

Literature search

We used several search methods to ensure a thorough examination of potential candidate reports. The number of records identified, included and excluded and the reasons for exclusions are shown in Fig. 3. The literature search covered a timeframe up to August 2022.

Multiple-database searches. To obtain relevant articles, we used specific keywords with wildcards, performing a combined search of the

following seven online databases: (1) PsycInfo, (2) Google Scholar, (3) MEDLINE, (4) PubMed, (5) ProQuest Dissertations and Theses Abstracts and Indexes: Social Sciences, (6) Communication Source and (7) Social Sciences Citation Index. We paired a series of keywords (that is, misinformation OR misbelief* OR false information OR belief perseverance OR continued influence) with two other series of keywords (that is, [retract* OR correct*] and [label* OR tag* OR flag*]). We expanded the sample of relevant reports by examining the reference lists of a systematic collection of review articles, book chapters and dissertations. The search yielded 2,882 studies.

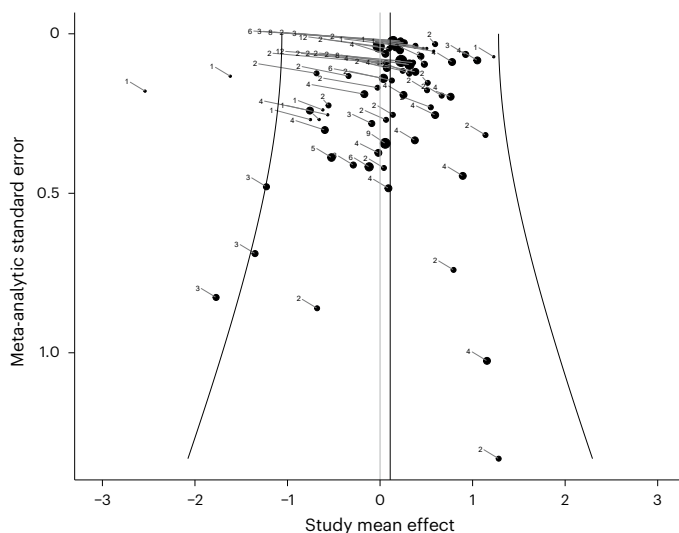


Fig. 4 | Study-funnel plot. Each dot represents a report in an article, and the numbers represent the number of effect sizes included in the estimation.

Other searches, personal contact and electronic platforms. By culling the reference lists of the review papers obtained through the database searches, we were able to identify eight additional articles. We also identified ten studies after contacting a list of researchers who have researched in this area. Additionally, we received materials from additional researchers after posting requests on online forums and e-mail list servers (for example, Society for Personality and Social Psychology). Finally, we searched the Open Science Framework (OSF) using the same set of keywords in early 2021 to obtain 258 unpublished and in-press datasets.

Criteria for inclusion/selection

We included (1) studies that assessed false claims concerning a scientific measurement procedure or scientific evidence (for examples of excluded reports due to measuring non-science-relevant misinformation, see Supplementary Table 2). For example, we included Vraga et al.'s⁸⁶ experiments studying logic-based and humour-based corrections for misinformation about climate change and the HPV vaccine. Another example of the included reports was Vijaykumar et al.'s⁸⁷ experiments that examined corrections for the inaccurate treatment effect of garlic to cure COVID-19. We also included Anderson et al.'s⁸⁸ experiments ostensibly evaluating the relation between firefighter performance and risk-seeking traits. Likewise, we included Greitemeyer's¹⁴ experiments studying the impact of a false link about the relation between the embodiment of height and pro-social behaviour.

Next, we used several eligibility criteria to select reports for inclusion in the meta-analysis, inspecting only studies from reports that were clearly or possibly experimental. Included studies (2) empirically measured participants' beliefs in or attitudes consistent with the misinformation addressed by the correction (for examples of excluded reports after evaluating the full text, see Supplementary Table 3). However, studies that included outcome measures of misinformation sharing (intentions)^{10,89–92}, the quality judgement of news sources⁹³, self-efficacy, openness to the messages⁹⁴ and whether participants responded to/ignored the corrections were excluded⁹⁵. The included studies (3) had a control or baseline group exposed to either no message, a neutral message or an unrelated message and (4) were eligible when the misinformation was initially asserted to be true or was known to participants before the study and was later corrected. All studies introduced a correction for the misinformation regardless of whether the misinformation was fictitious (for example, ref. 88) and known to be familiar to the public (for example, refs. 16,95). However, studies

that described the initial information as hypothetical or uncertain (for example, ref. 96) or as an accusation of scientific misconduct (for example, ref. 97) were excluded. The final dataset of this meta-analysis included 75 research reports ($N = 53,320$). Reports with multiple experiments and/or experimental groups often contributed more than one effect size.

Estimation of the effect sizes

We used Hedges' d as the metric for effect-size estimation in our meta-analysis. Hedges' d and Hedges' g pool variances are based on the assumption of equal population variances, and both metrics can be interpreted in the same way⁹⁸. However, Hedges' d includes an adjustment factor j , that is, $1 - 3/(4 \times n - 1)$, for each sample, and in turn, reduces the positive bias for a small sample that is common in experimental studies^{99,100}. Two trained raters first decided whether the report was in a between- or within-subjects design and then selected corresponding means and s.d. from different groups or conditions to compute Hedges' d in accordance with the formulas outlined by Borenstein et al.¹⁰⁰. All d statistics are in the same normalized units regardless of whether they derive from between- or within-subjects designs¹⁰⁰. If a particular study did not report any of these statistics, the rater recorded other relevant statistics, such as F ratios or t values, and then obtained Hedges' d on the basis of the step-by-step workflow as specified in Lakens's effect-size calculation spreadsheet¹⁰¹. Given the inclusion of different experimental designs in this meta-analysis, the d obtained from the reports compared different means as explained presently. We obtained effect sizes for misinformation and debunking. Debunking effects combined correction and the reverse of misinformation-persistence effects. We followed different procedures to calculate the variances of effect sizes. In particular, calculations of the between-subjects variances followed Hedges and Olkin's⁹⁸ procedures, and calculations of the within-subjects variances followed Morris's¹⁰² procedures with a correlation set at 0.5 between repeated measures.

Within-subjects design. We first illustrate the effects of correction and misinformation persistence using a within-subjects design. Imagine that participants were recruited for an experiment with a pre-test–post-test design and that they provided a rating on a belief or attitude measure from 0 to 9 to indicate their belief or attitude both before (pre-test) and after the experimental manipulations of misinformation (post-test 1) and correction (post-test 2). The comparisons among the ratings for the pre-test and post-test 1 generate a misinformation effect. The comparisons among the ratings for post-tests 1 and 2 generate a correction effect, whereas the comparisons between the ratings for the pre-test and post-test 2 allow us to calculate a misinformation-persistence effect size. For example, imagine that participants gave a rating of 1 at the pre-test, a rating of 9 after the receipt of the misinformation (post-test 1) and then a rating of 6 after correcting the misinformation (post-test 2). Then, the misinformation effect is the difference between the ratings at post-test 1 and at the pre-test (that is, $9 - 1 = 8$); the correction effect is the difference between the ratings at post-tests 1 and 2 (that is, $9 - 6 = 3$); and the misinformation-persistence effect is the difference between the ratings at post-test 2 and the pre-test (that is, $6 - 1 = 5$). When pre-test ratings were unavailable, we used the ratings obtained from a control group as the baseline for comparisons with the ratings at post-tests 1 and 2 using between-subjects procedures.

Between-subjects design. Imagine now a between-subjects design with three groups of participants. Imagine also that participants in the misinformation group received only misinformation, participants in the correction group received the misinformation and subsequently a correction, and participants in the control group received either no information or information on an unrelated topic. Now consider that

all participants provided a rating on a belief or attitude measure from 0 to 9. Participants in the misinformation group gave an average rating of 9, participants in the control group gave an average rating of 1, and participants in the correction group gave an average rating of 6. In these circumstances, the differences between the misinformation group and the control group constitute the misinformation effect (that is, $9 - 1 = 8$); the differences between the misinformation-only group and the correction group constitute the correction effect (that is, $9 - 6 = 3$); and the differences between the correction group and the control group constitute the misinformation-persistence effect (that is, $6 - 1 = 5$). Accordingly, d values greater than 0 for later persistence of the misinformation indicate that recipients of the corrections showed more misinformation persistence than participants in the comparison group (for example, the control group).

Coding of moderators

Two authors and a trained research assistant coded for four theoretical moderators, including the variables of interest: (1) whether the misinformation was about negative or neutral topics, (2) the level of detail of the correction messages, (3) the attitudinal congeniality of the correction and (4) the issue polarization and four other control factors, including (5) whether the misinformation was about politics, health, environment and others, (6) whether the misinformation was about fictitious or real issues, (7) the likelihood of familiarity with the topic, and (8) the use of in-person correction^{9,46,103–119}. Two rounds of coding of all variables contained about 14% of the reports, and the coding reached an adequate agreement (Krippendorff's α : mean 0.99, s.d. 0.01 and Cohen's κ : mean 0.95, s.d. 0.09). Further, the coders resolved all disagreements by discussion and consultation with another author. Table 4 summarizes effect sizes for each level of all categorical moderators and the number of experimental conditions coded for each level. We next detailed definitions and examples of all moderators.

Misinformation factors. Negativity of the misinformation. We coded whether the misinformation topic was negative or neutral. As a first example, Guenther and Alické's¹¹⁰ misinformation about failure feedback on an alleged test of mental acuity to measure a fundamental aspect of intelligence was coded as negative (that is, a score of 2) because of the potential to induce sadness or anxiety. As another example, Anderson's¹⁸ experiments included misinformation about whether risk-seeking or risk-averse firefighter trainees performed better at their job. This misinformation topic was coded as neutral, receiving a score of 1. Only three reports from two studies contained positive misinformation topics. In two, participants received flattering feedback on their cognitive ability based on a task-performance task¹¹¹ and on a word-identification task supposedly linked to intelligence¹¹⁰. We thus analysed the data excluding these three reports.

Correction factors. Level of detail of the correction messages. The two raters also coded whether the correction simply labelled the initial information as incorrect (1, succinct) or provided detailed information (2, detailed). For example, a detailed correction message (that is, the author realized that some of the facts in the reading were not true) describing why the initial misinformation (that is, the facts of the reading were 'mixed up' with facts of a fictional story also to be published) was incorrect was considered as a detailed correction message (a score of 2) (ref. 112).

Recipient factors. Attitudinal congeniality of the correction. This variable captured whether participants had any pre-existing attitudes relative to the position advocated in the correction message. For example, Ecker and Ang's¹¹³ experiment 1 disseminated different corrections to participants who reported being left-wing. Here the conditions with consistent partisan information (for example, Labour supporters receiving left-wing correction) were coded as 1. The conditions with either inconsistent partisan information (for example, Liberal

supporters receiving left-wing correction) or non-partisan information were coded as -1.

Issue polarization. We coded whether the topic was associated with disagreement between opposing groups in the country where the experiment was carried out (polarizing, 1; non-polarizing, -1).

Control factors. Domains of the misinformation. We coded whether the misinformation was about politics, health, environment and others (politics, 1; health, 2; environment, 3; others, -1) on the basis of the misinformation included in the reports, regardless of whether it was politicized in the real world. For example, the alleged measles, mumps and rubella vaccines-autism link was coded as concerning health (2) and misconceptions about genetically modified foods were coded as concerning the environment (3). Reports with different misinformation (and possibly different domains) were coded as separate records whenever possible, for example⁸⁶. As only four reports ($k = 10$) had misinformation about multiple domains, we decided not to include all possible combinations of domains as separate coding options.

Fictitious issue. We coded whether the claim was fictitious (1) or real (-1). For example, the alleged link between Zika virus vaccines and epilepsy was never true, receiving no scientific support (1), whereas there was scientific support (even minimal) for hydroxychloroquine to be effective against COVID-19 (-1).

Likely familiarity with the topic. We coded for whether the topic used in the experiment had circulated in the real world. For example, Ecker¹⁰⁸ presented vaccine misinformation concerning the link between the measles, mumps and rubella vaccine and autism to UK participants, a topic of wide dissemination in the United Kingdom. This study was coded as 1. In contrast, Sherman and Kim's¹¹⁴ experiments used the topic of the associations between Chinese characters and English meanings, which was coded as -1 (that is, likely unfamiliar topic).

In-person correction. We next coded whether the correction was given in person or not. As an example of in-person delivery, Golding et al.'s¹¹⁵ correction involved an experimenter telling research participants in the lab to disregard initial misinformation, and it was coded as in person (that is, a score of 1). In contrast, Sherman and Kim's¹¹⁴ experiments presented the experimental materials using computer software and were coded as not in person (that is, a score of -1).

Report and methodological characteristics. Geographical location of the study sample. We coded whether participants were self-reported as from the United States or other countries (United States, 1; other countries, 2).

Lab context. We coded whether the experiment was carried out in the lab (a score of 1) or online (a score of 2).

Methods of effect-size calculation. We recorded whether the effect size of the report stemmed from a between-subjects (a score of 1) or within-subjects (a score of 2) design.

Source type. We coded the report's publication category (published article, 1; working paper, 5; dissertation/thesis, 6; unpublished data, 7).

Bias analysis

Examining variability and bias is critical in meta-analysis because much research is affected by both high variability and bias. We adopted the diagnostic procedures proposed by Viechtbauer and Cheung¹¹⁶ to detect influential cases, and six debunking effects were identified as outliers, $d < -2.37$ or $d > 2.61$. Because outliers and influential cases may represent random noise or reflect systematic heterogeneity as a

function of specific moderators, we performed six bias tests to assess publication/inclusion biases for effect sizes with and without the outliers. Figure 4 shows the study-level funnel plot (for the study-level forest plot, see Supplementary Information).

We performed bias tests to assess publication/inclusion biases for effect sizes with and without outliers^{65,117–124}. Overall, the bias tests showed no consistent results regarding the presence of any bias in the dataset (Table 2). Table 2 presents a consistent pattern of the results of bias analyses between the datasets with and without the outliers. The rank correlation test and the log-likelihood ratio tests method⁶⁵, showed the possibility of bias in the dataset ($P < 0.05$). In contrast, the trim-and-fill method, the meta-regression analyses of publication type, the PET-PEESE (precision-effect test and precision-effect estimate with standard errors) test, the three-parameter selection method¹²³, the robust Bayesian meta-analysis (RoBMA; $BF_{10} < 1$)¹²⁴, and the P -uniform test showed no evidence of the presence of bias.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are openly available in OSF at <https://osf.io/vkygw/>.

Code availability

All code for data analyses associated with the current submission is available at <https://osf.io/vkygw/>. Any updates will also be published in OSF.

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Author contributions

D.A. initiated the project, and M.S.C. supervised the project. Both M.S.C. and D.A. contributed to the theoretical formalism, developed the coding scheme and performed the coding reliability. M.S.C. took the lead in the data curation, preparing the analytical plan and performing the analytic calculations. Both M.S.C. and D.A. discussed the results and contributed to the final version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Software and code

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Study description

This study was a meta-analysis of a large body of experimental evidence comprising 61 published experiments, five working papers, two theses, and seven unpublished datasets (total number of reports = 75; k of effect sizes = 245). The data are quantitative.

Research sample

The reports were obtained from searches of online databases, e.g., PsycInfo. We included all the samples as reported in the included reports. The number of participants in the synthesis ranged from 7 to 3142 their mean age was 37 years old (SD = 10.49), and about 60 percent of them were female. Participants included university students, graduate students, and adults from the community recruited via online survey platforms, such as Amazon Mechanical Turk and Prolific.

Sampling strategy

The sample size was the total number of participants reported in the included reports. No additional sampling strategy was used.

Data collection

We used specific keywords with wildcards, performing a combined search of the following seven online databases: (a) PsycInfo, (b) Google Scholar, (c) MEDLINE, (d) PubMed, (e) ProQuest Dissertations and Theses Abstracts and Indexes: Social Sciences, (f) Communication Source, and (g) Social Sciences Citation Index. We paired a series of keywords (i.e., misinformation OR misbelief* OR false information OR belief perseverance OR continued influence) with two other series of keywords (i.e., [retract* OR correct*] and [label* OR tag* OR flag*]). We expanded the sample of relevant reports by examining the reference lists of a systematic collection of review articles, book chapters, and dissertations.

By culling the reference lists of the review papers obtained through the database searches, we were able to identify eight additional articles. We also identified ten studies after contacting a list of researchers who have researched in this area but have no awareness of the research questions and hypotheses. Additionally, we received materials from additional researchers after posting requests on online forums and e-mail list servers (e.g., Society for Personality and Social Psychology). Finally, we searched the Open Science Framework (OSF) using the same set of keywords in early 2021 to obtain 258 unpublished and in-press data sets.

Authors and coders were blinded to the experimental conditional and hypotheses of the included reports. They were not blinded to the meta-analysis's research questions as they were preregistered at OSF <https://osf.io/vkygw>

Timing

We did not set the start date of the literature search and the end date was on August 31 2022.

Data exclusions

We included (a) studies that assessed false claims concerning a scientific measurement procedure or scientific evidence. We used several eligibility criteria to select reports for inclusion in the meta-analysis, inspecting only studies from reports that were clearly or possibly experimental. Included studies (b) empirically measured participants' beliefs in or attitudes consistent with the misinformation addressed by the correction (see Table S3 for examples of excluded reports after evaluating the full text). However, studies that included outcome measures of misinformation sharing (intentions) (Arechar et al., 2022; Jahanbakhsh et al., 2021; Pennycook et al., 2020, 2021; Sirin et al., 2021), the quality judgment of news sources (Pennycook & Rand, 2019a), self-efficacy, openness to the messages (Gesser-Edelsburg et al., 2018), and whether participants responded to/ignored the corrections were excluded (Mosleh et al., 2022). The included studies (c) had a control or baseline group exposed to either no message, a neutral message, or an unrelated message and (d) were eligible when the misinformation was initially asserted to be true or was known to participants prior to the study and was later corrected. All studies introduced a correction for the misinformation regardless of whether the misinformation was fictitious (e.g., Anderson et al., 1980) and known to be familiar to the public (e.g., Andrews, 2021; Yousuf et al., 2021). However, studies that described the initial information as hypothetical or uncertain (e.g., Koller, 1993) or as an accusation of scientific misconduct (e.g., Greitemeyer & Sagioglou, 2015) were excluded. The coding protocol and exclusion criteria were preregistered in mid-2022 at OSF <https://osf.io/vkygw>.

Non-participation

No participants were involved in this study

Randomization

Randomization was not applicable to this study as this is a meta-analysis.

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